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Approved by AICTE, New Delhi | Recognized by D.T.E. Govt. of Maharashtra - Code : 6456
Affiliated to the M.S.B.T.E. Mumbai - Code : 1227

A PROJECT

REPORT

ON

IoT BASED SMART PAPERWEIGHT

SUBMITED TO

MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE AWARD OF

DIPLOMA IN COMPUTER ENGINEERING

BY

MONAL MESHRAM

UNDER THE GUIDANCE OF

PROF. V. B. JADHAV

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PROJECT REPORT ON

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BY

MR. HRISHIKESH MEHTRE(CO01)

MR. AMIT TAVASKAR(CO06)

MISS. MONAL MESHRAM(CO28)

MR. RUSHIKESH KIRDE(CO38)

IN PARTIAL FULFILLMENT OF: -

DIPLOMA IN COMPUTER ENGINEERING

MSBTE

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DEPARTMENT OF COMPUTER ENGINEERING

K.E. SOCIETY'S RAJARAMBAPU INTITUTE OF TECHNOLOGY'S (POLYTECHNIC)

LOHEGAON, PUNE-411047

(2024-2025)

CERTIFICATE

This is to certify that the report entitled

IoT BASED SMART PAPERWEIGHT

Submitted by

MONAL SHARAD MESHRAM

To MSBTE towards partial fulfilment of the diploma of

COMPUTER ENGINEERING

MSBTE

Has been satisfactorily completed under the guidance of

PROF. V. B. JADHAV

During the academic year 2023-2024

Prof. V. B. Jadhav	**Prof. V. B. Jadhav**	**Examiner**	**Dr. K. H. Munde**
(Project Guide)	**(Head of Department)**	**(External)**	**(Principal)**

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It gives us great pleasure to present this report on **IoT BASED SMART PAPER WEIGHT** towards partial fulfilment of Diploma in Computer Engineering course of Maharashtra State Board of Technical Education (MSBTE).

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My special thanks to Head OF Computer Engineering Department and Principal of K.E.S. Polytechnic for their cooperation and support.

I am also thankful to all faculties of Computer Engineering Department for their help.

ABSTRACT

In the era of smart automation, everyday objects are being enhanced with intelligent functionalities. This paper presents a smart paperweight that goes beyond its traditional role by integrating gesture-based control for home and office appliances. This innovative device can turn on/off lights and fans and control fan speed using rotational gestures.

The smart paperweight is equipped with an inertial measurement unit (IMU) sensor to detect clockwise and anti-clockwise rotations. When rotated clockwise, it increases the fan speed, while an anti-clockwise rotation decreases the speed and eventually turns the fan off. Additionally, a simple tap or press on the paperweight can turn the lights on or off. The device is powered by a microcontroller (such as ESP32 or Arduino) and communicates wirelessly via Wi-Fi or Bluetooth to connected appliances.

This hands-free control system enhances convenience, energy efficiency, and user experience, making it ideal for smart homes, offices, and study spaces. By combining IoT and gesture-based control, the smart paperweight redefines the functionality of a simple desk accessory, transforming it into an intuitive and modern automation tool.

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INTRODUCTION

With the rapid advancement of the Internet of Things (IoT) and smart automation, traditional objects are being reimagined to serve multiple purposes. A paperweight, which is typically used to keep papers in place, can be enhanced with smart technology to offer additional functionalities. This project introduces a smart paperweight that not only serves its conventional purpose but also acts as a gesture-based controller for electrical appliances like lights and fans.

The smart paperweight integrates motion sensors and microcontrollers to detect user gestures. By rotating it clockwise, users can turn on the fan and adjust its speed, while an anti-clockwise rotation gradually decreases the speed and eventually turns the fan off. Additionally, a single tap or press on the paperweight can control lights, adding further convenience to daily tasks. The device communicates wirelessly via Wi-Fi or Bluetooth, allowing seamless interaction with connected appliances.

This innovation eliminates the need for physical switches, making it a hands-free, intuitive, and energy-efficient solution for smart homes, offices, and study spaces. By combining IoT, gesture recognition, and automation, the smart paperweight not only enhances user experience but also promotes modern and efficient control of everyday appliances.

The IoT-based smart paperweight is an innovative device that combines traditional functionality with smart automation. Designed to control lights and fans through gesture-based movements, this smart paperweight enhances convenience and energy efficiency. By rotating it clockwise, the fan turns on and its speed increases, while an anti-clockwise rotation decreases the speed and eventually turns it off. Additionally, a tap or press can turn the lights on or off.

Equipped with motion sensors, a microcontroller, and wireless connectivity (Wi-Fi/Bluetooth), this device eliminates the need for physical switches, offering a hands-free and intuitive control mechanism. Ideal for smart homes, offices, and study spaces, the smart paperweight redefines how users interact with everyday appliances, making automation more accessible and user-friendly.

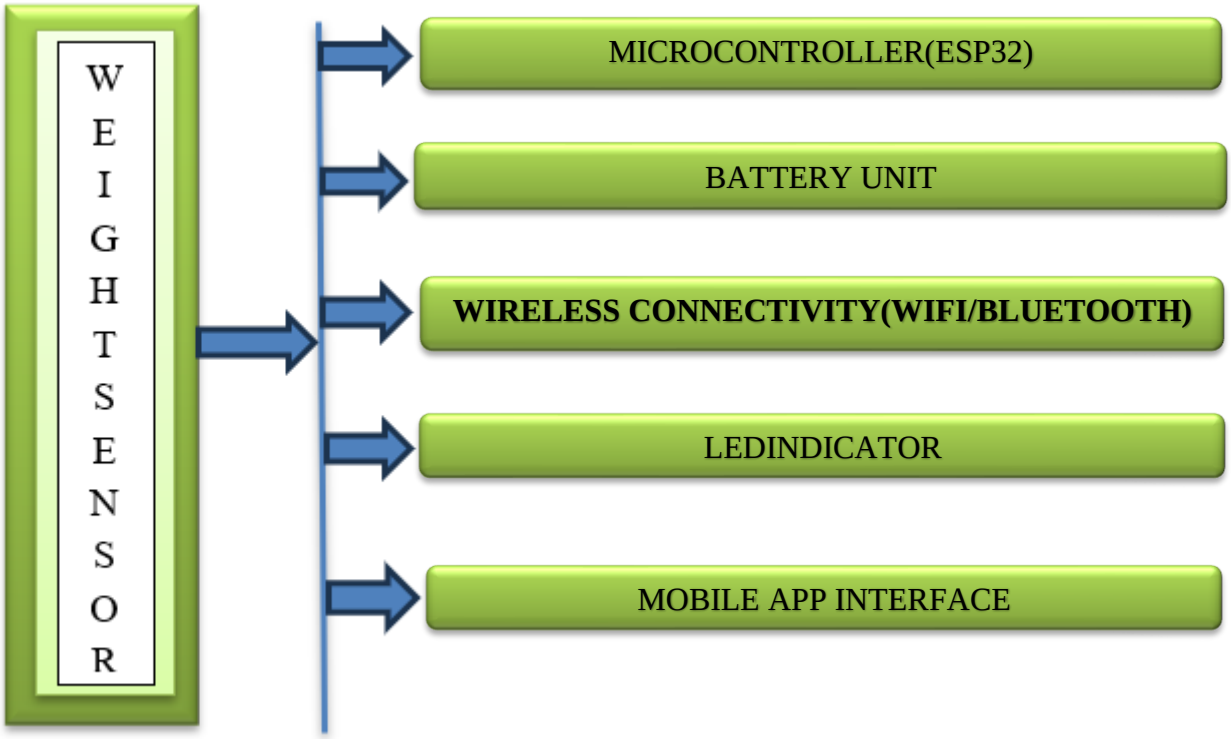


FIG:-MODULE OF SYSTEM

1. ARDUINO IDE:

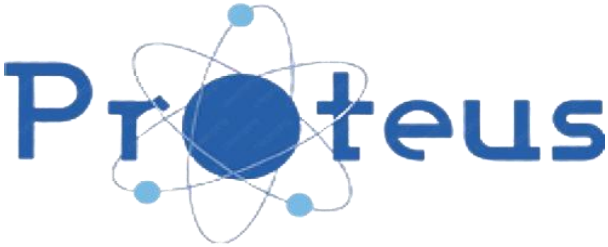


The Arduino IDE (Integrated Development Environment) is a software application used to write, compile, and upload code to Arduino microcontrollers. It is designed to be simple and beginner-friendly, making it an ideal choice for hobbyists, students, and professionals who want to develop projects using Arduino boards. The IDE is based on a C++-derived language with additional libraries that simplify hardware interaction, such as controlling LEDs, sensors, motors, and displays.

Arduino IDE provides an intuitive code editor where users can write and edit programs called sketches. It features essential tools such as a Serial Monitor for debugging, a Library Manager for adding pre-written code libraries, and a Board Manager that allows users to work with different Arduino-compatible boards. The software also includes a Serial Plotter to visualize real-time data coming from sensors or other inputs.

There are two major versions of the Arduino IDE. The Arduino IDE 1.x is the classic version with a simple interface and basic functionality, whereas the Arduino IDE 2.x introduces improvements like auto-completion, real-time error detection, dark mode, and an integrated debugger for supported boards. The newer version also offers faster compilation times and a modern user interface, making it more efficient for advanced users.

2. PROTEUS:



Proteus is a powerful simulation and design software used primarily for electronic circuit design, PCB (Printed Circuit Board) layout, and microcontroller simulation. Developed by Labcenter Electronics, it is widely used by engineers, students, and hobbyists for designing, testing, and verifying circuits before implementing them in hardware. One of its most notable features is the ability to simulate microcontrollers such as Arduino, PIC, AVR, and 8051, allowing users to run embedded code alongside their circuit designs.

The software includes several integrated tools, such as Proteus ISIS (Intelligent Schematic Input System) for designing circuit schematics and Proteus ARES (Advanced Routing and Editing Software) for PCB design. With these tools, users can create, modify, and test circuits in a virtual environment, reducing the need for physical prototyping and saving both time and resources. Proteus supports a vast library of electronic components, including resistors, capacitors, transistors, ICs, sensors, and microcontrollers, making it an all-in-one solution for circuit design.

One of the major advantages of Proteus is its real-time simulation capabilities. Users can write and upload embedded C or assembly code to microcontrollers within the software and observe how the circuit behaves. This feature is particularly useful for debugging and testing embedded systems, as it allows engineers to detect and fix issues before implementing the design on physical hardware. Proteus also includes a virtual oscilloscope, logic analyzer, and signal generator, enabling users to analyze circuit performance in detail.

3. COREL DRAW:



CorelDRAW

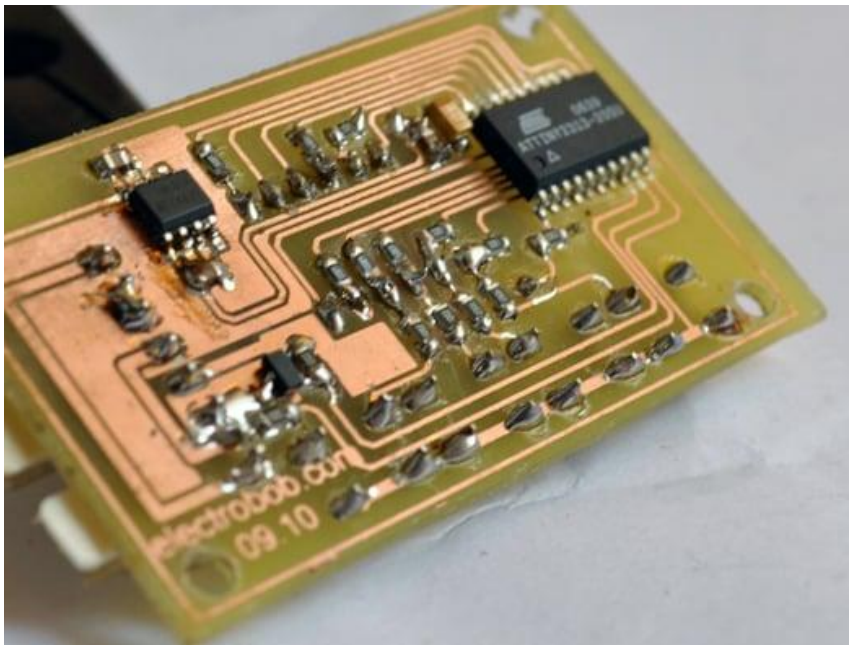
CorelDRAW is a powerful vector graphics design software developed by Corel Corporation, widely used for creating illustrations, logos, brochures, business cards, advertisements, and other graphic design projects. It is particularly popular among professionals in the fields of graphic design, marketing, advertising, and print media due to its intuitive interface and extensive set of design tools.

One of CorelDRAW's key strengths is its vector-based design capability, which allows users to create scalable graphics that do not lose quality when resized. Unlike raster images, which are made up of pixels, vector graphics are composed of mathematical equations, enabling sharp and high-quality output for both digital and print media. CorelDRAW supports multiple file formats, including AI, EPS, PDF, SVG, and PSD, making it highly compatible with other design software like Adobe Illustrator and Photoshop.

CorelDRAW comes with a range of powerful features, including Bezier and Pen tools for precise drawing, Shape and Text tools for creative typography, and LiveSketch, an innovative AI-powered tool that converts freehand sketches into precise vector curves. Additionally, the software includes photo-editing capabilities, advanced color management, and page layout tools, making it a versatile solution for designers.

CorelDRAW is available as a standalone product or as part of the CorelDRAW Graphics Suite, which includes additional tools like Corel PHOTO-PAINT (a photo-editing application), Corel Font Manager for managing fonts, and PowerTRACE, a tool that converts raster images into editable vector graphics. The software is compatible with Windows and macOS and offers both a one-time purchase option and a subscription model, catering to different user needs.

4. PRINTED CIRCUIT BOARD (PCB):

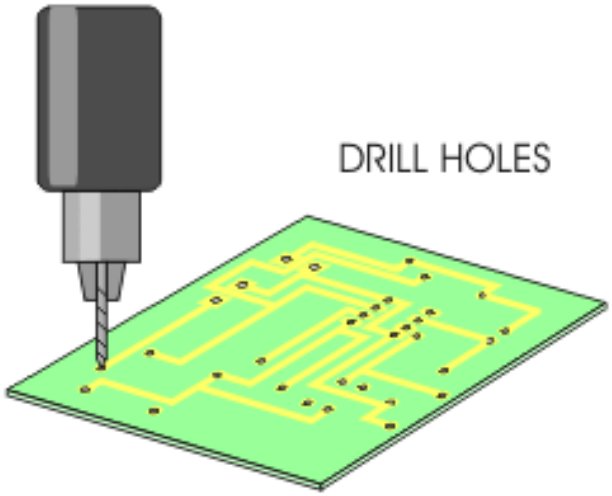


A Printed Circuit Board (PCB) is a crucial component in modern electronics, serving as the foundation for assembling and connecting electronic components. PCBs are used in almost all electronic devices, from smartphones and computers to industrial machinery and medical equipment. They provide a structured platform where components such as resistors, capacitors, microcontrollers, and integrated circuits (ICs) are soldered and interconnected through conductive pathways.

PCBs are made from non-conductive materials like fiberglass (FR4), with thin layers of conductive copper traces etched onto the surface. These traces act as electrical pathways that connect different components, eliminating the need for bulky wiring. The design of a PCB can range from single-layer (basic circuits) to multi-layered (complex systems like computer motherboards), depending on the application's complexity.

The PCB manufacturing process typically involves designing the schematic, creating a layout using PCB design software (such as Proteus, Eagle, Altium Designer, or KiCad), fabricating the board, and finally assembling and soldering components. The etching and printing process ensures that only the required copper traces remain on the board, allowing for precise and efficient circuit connections.

5. DRILLING:



PCB drilling is a crucial step in printed circuit board (PCB) manufacturing, where precise holes are drilled for component placement, electrical connections, and mechanical support. These holes, known as plated through-holes (PTH), non-plated through-holes (NPTH), and vias (through, blind, or buried), enable proper circuit functionality and multi-layer connectivity. The drilling process must be highly precise, as incorrect hole sizes or misalignment can lead to circuit failures.

The drilling process is typically carried out using CNC (Computer Numerical Control) machines with carbide or diamond-coated drill bits to ensure accuracy and durability. Mechanical drilling is the most common method, while laser drilling is used for microvias in high-density interconnect (HDI) PCBs, allowing for miniaturization and improved performance. After drilling, holes are cleaned and plated (if necessary) to establish electrical connections between layers.

With advancements in automation and laser technology, PCB drilling has become more efficient, allowing for the production of smaller, more complex, and high-performance circuit boards. These innovations support industries such as aerospace, automotive, telecommunications, and consumer electronics, where precision and reliability are critical. Proper drilling ensures that the final PCB meets design specifications and functions optimally in its intended application.

6. SOLDERING:



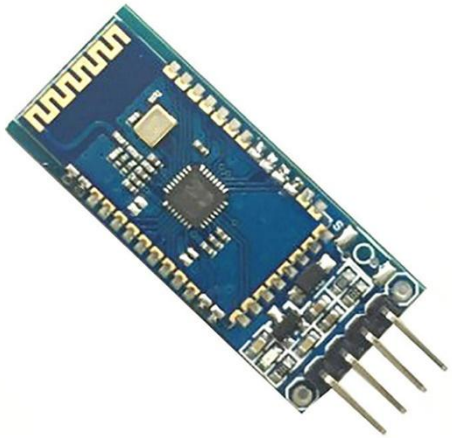
Soldering is a fundamental process in electronics and metalwork, used to join two or more metal components by melting a filler metal, called solder, which solidifies to create a strong electrical and mechanical bond. It is widely used in PCB assembly, electrical repairs, plumbing, and metal fabrication. Soldering provides a durable and conductive connection while ensuring minimal damage to the components being joined.

There are three main types of soldering: soft soldering, hard soldering, and brazing. Soft soldering is the most common method for electronics, using a low-melting-point alloy (typically tin-lead or lead-free solder) and a soldering iron. Hard soldering, such as silver soldering, uses higher temperatures and stronger joints, often applied in jewelry and metalwork. Brazing, similar to welding, uses a high-temperature filler metal and is mainly used for joining structural metals.

In electronics, soldering is essential for connecting components to PCBs. The process involves cleaning the surfaces, heating the joint with a soldering iron, applying solder, and allowing it to cool for a secure bond. Flux, a chemical agent, is used to remove oxidation and improve solder flow. Common soldering techniques include through-hole soldering, surface-mount soldering (SMT), and reflow soldering, which is used for automated PCB assembly.

Proper soldering requires skill and precision to avoid defects like cold joints, solder bridges, and excess solder, which can cause poor conductivity or short circuits. Advanced soldering methods, such as hot air rework and wave soldering, are used in large-scale manufacturing. As technology advances, lead-free soldering has become the industry standard due to environmental and health concerns, replacing traditional lead-based solder with tin-silver-copper alloys.

7. BLUETOOTH:



Bluetooth is a wireless communication technology that enables short-range data exchange between devices using radio waves. It operates on the 2.4 GHz frequency band and is widely used in smartphones, computers, headphones, speakers, smartwatches, IoT devices, and automotive systems. Bluetooth eliminates the need for physical cables, making it convenient for wireless audio streaming, file transfers, and peripheral connections like keyboards and mice.

Bluetooth technology works by establishing a paired connection between two devices using a process called device discovery and pairing. Once connected, devices communicate using low-power signals over short distances, typically up to 10 meters (33 feet) for standard Bluetooth and up to 100 meters (328 feet) for long-range versions. The technology has evolved over time, with different versions such as Bluetooth 4.0, 5.0, and 5.3, each improving on range, speed, energy efficiency, and security.

One of the key advancements in Bluetooth is Bluetooth Low Energy (BLE), introduced in Bluetooth 4.0, which significantly reduces power consumption. BLE is commonly used in wearable devices, fitness trackers, medical sensors, and smart home applications, where long battery life is essential. Bluetooth 5.0 and later versions further improve data transfer speed, connection stability, and multi-device support, making them ideal for modern wireless applications, including IoT and smart home automation.

8. BATTERY:

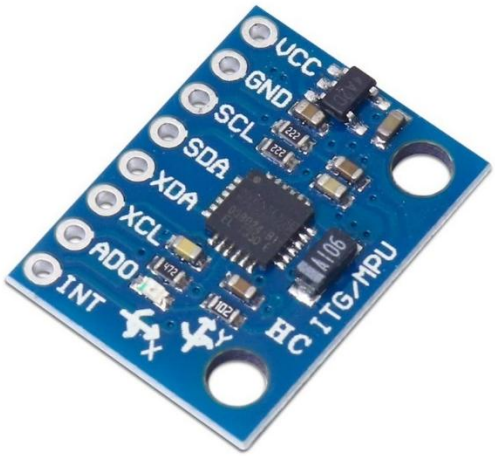


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9. GYRO:



A **gyroscope (gyro)** is a sensor that measures the angular velocity or orientation of an object. It detects rotational motion around one or more axes, such as pitch (forward/backward tilt), yaw (left/right rotation), and roll (side tilt). In IoT-based projects, gyroscopes are essential for tracking motion, maintaining balance, and determining the direction of movement. They are commonly integrated into an **Inertial Measurement Unit (IMU)** along with accelerometers for more accurate motion detection.

In the world of **IoT (Internet of Things)**, gyroscopes are widely used in devices like smart wearables, smartphones, drones, and robots. For example, in wearable fitness trackers, a gyro can detect movements and orientation changes during workouts. In drones and autonomous vehicles, gyros help with navigation, stability, and orientation control. They are also used in health monitoring devices to detect falls or abnormal movements, especially for elderly care systems.

Popular gyroscope modules used in IoT projects include the **MPU-6050** (which combines a 3-axis gyroscope and 3-axis accelerometer), the **L3G4200D**, and the **MPU-9250**, which offers a full 9 degrees of freedom by including a magnetometer as well. These modules communicate with microcontrollers like **Arduino**, **ESP32**, or **Raspberry Pi** using I2C or SPI communication protocols, making them easy to integrate into IoT systems.

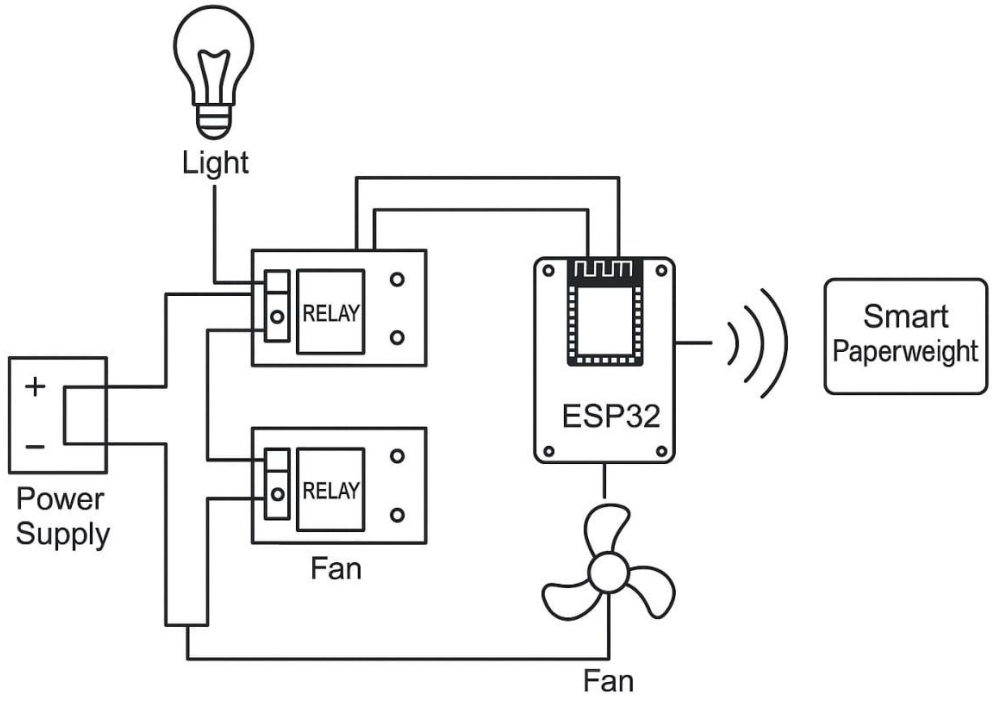
Overall, gyroscopes add significant value to IoT projects by enabling motion tracking, gesture recognition, and orientation sensing. This makes them ideal for a wide range of smart applications such as self-balancing robots, gesture-controlled appliances, and smart helmets for safety monitoring. By accurately capturing rotational movement, gyros enhance the functionality and responsiveness of IoT devices.

PROPOSED SYSTEM

The proposed system is an IoT-enabled Smart Paperweight that not only serves as a traditional desk accessory but also functions as a smart monitoring device. It is designed to measure weight placed on it using a load cell sensor and detect any unauthorized movement. The data collected is processed through a microcontroller (ESP8266/ESP32) and transmitted wirelessly to a smartphone or IoT dashboard for real-time monitoring.

This smart paperweight integrates a HX711 amplifier module to read precise weight measurements from the load cell and forwards the data to the microcontroller. The system is programmed to continuously monitor changes in weight or displacement, which can indicate if any important document or item has been removed. If any significant change is detected, it can trigger an alert or update the user via a mobile app using Wi-Fi-based IoT connectivity (e.g., Blynk or ThingSpeak).

Additionally, the device can be expanded with other sensors such as gyroscope (for motion detection) or temperature sensors (for environmental monitoring), making it a multi-functional desk gadget. The proposed system thus combines simplicity and functionality to deliver a smart security solution for office or personal use, ensuring sensitive documents are always under surveillance.



LITERATURE SURVEY

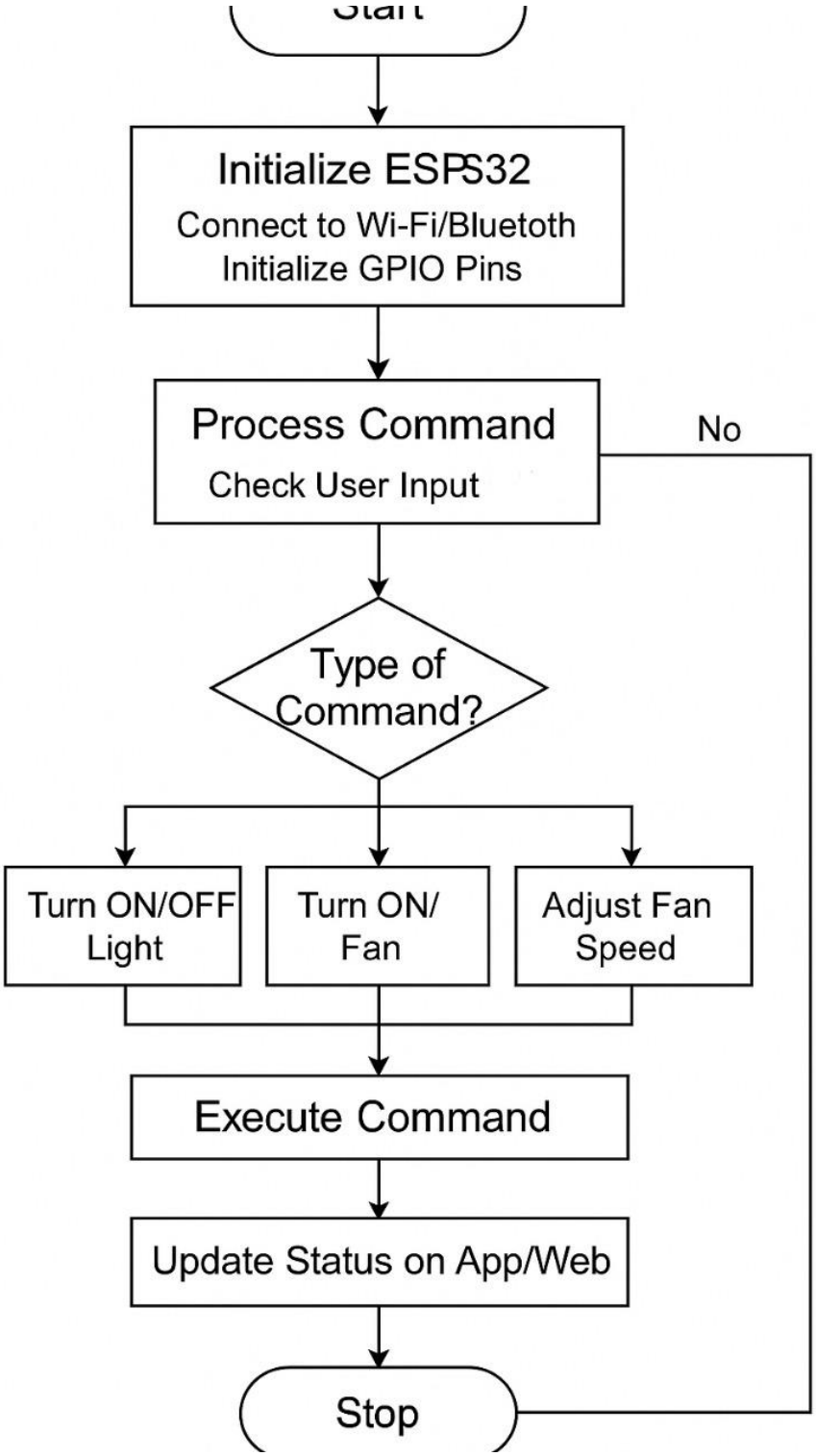
A literature survey on IoT-based smart paperweights reveals a growing trend in smart home automation, where everyday objects are embedded with IoT capabilities for enhanced functionality. Previous research on IoT-enabled devices highlights the importance of integrating sensors, wireless communication, and automation to improve energy efficiency and user convenience. Studies have shown that temperature and light-controlled systems significantly reduce energy wastage by automating appliance usage based on environmental conditions.

Existing IoT-based home automation systems, such as smart lighting and climate control solutions, utilize Wi-Fi, Bluetooth, and cloud-based services to enable remote access and real-time monitoring. Research also emphasizes the role of ESP8266/ESP32 microcontrollers in creating low-cost, efficient IoT systems. Moreover, advancements in Bluetooth Low Energy (BLE) and MQTT protocols have improved the reliability of such systems, allowing seamless device communication.

Studies on fan speed control mechanisms suggest that PWM (Pulse Width Modulation) and MOSFET drivers are effective for adjusting speed without excessive power consumption. Additionally, integrating voice-controlled assistants like Google Assistant and Alexa further enhances accessibility and user experience.

By combining insights from these studies, the proposed IoT-based smart paperweight brings together automation, remote accessibility, and energy efficiency, making it a practical solution for modern smart homes and offices.

FLOW CHART :-


Start
Initialize ESP32
Connect to Wi-Fi/Bluetoth
Initialize GPIO Pins
Process Command
Check User Input
No
Type of Command?
Turn ON/OFF Light
Turn ON/ Fan
Adjust Fan Speed
Execute Command
Update Status on App/Web
Stop

ARDUINO CODE :-

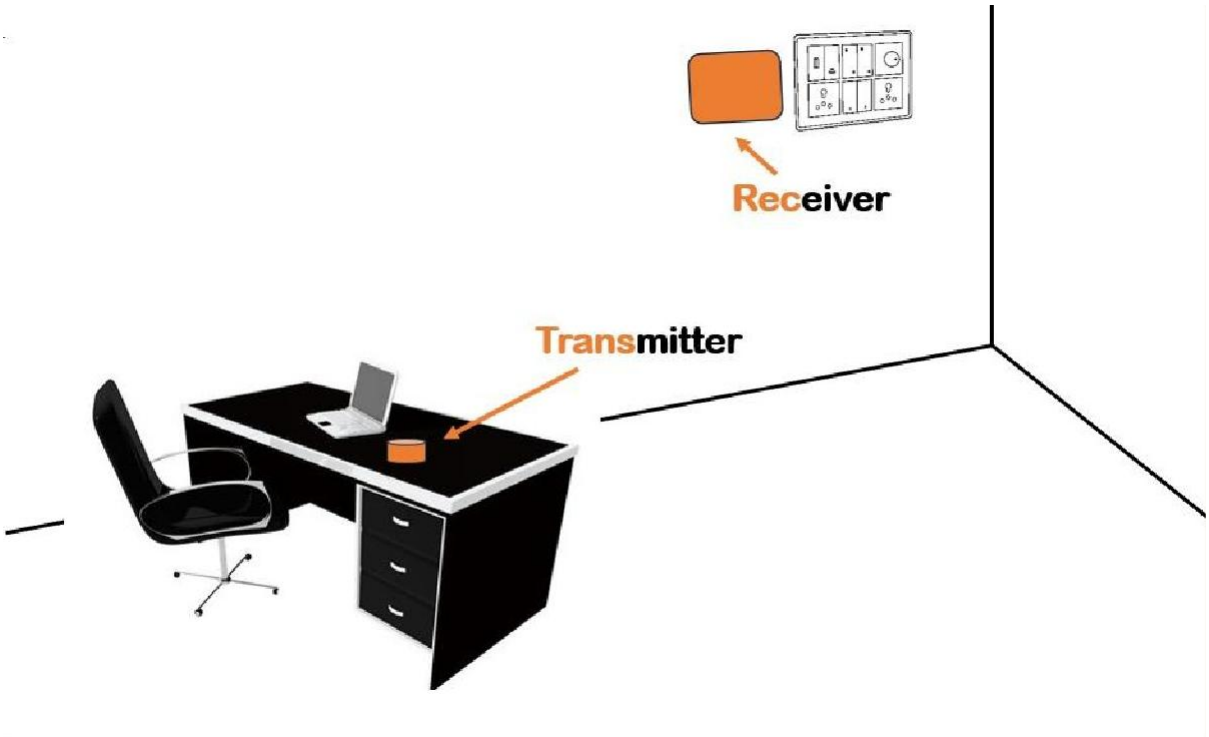
```
#define RELAY_PIN 23  // GPIO pin connected to the relay

void setup() {
  Serial.begin(115200);          // Start Serial communication
  pinMode(RELAY_PIN, OUTPUT);     // Set the relay pin as output
  digitalWrite(RELAY_PIN, LOW);   // Initially keep light OFF
  Serial.println("ESP32 Ready. Type 'on' or 'off' to control the light.");
}

void loop() {
  if (Serial.available() > 0) {
    String command = Serial.readStringUntil('\n');
    command.trim();  // remove any whitespace or new line characters

    if (command == "on") {
      digitalWrite(RELAY_PIN, HIGH);  // Turn ON the light
      Serial.println("Light turned ON");
    }
    else if (command == "off") {
      digitalWrite(RELAY_PIN, LOW);   // Turn OFF the light
      Serial.println("Light turned OFF");
    }
    else {
      Serial.println("Invalid command. Use 'on' or 'off'.");
    }
  }
}
```

OVERALL PROJECT DIAGRAM



PROPOSED WORK

The proposed project aims to develop a compact and innovative IoT-based Smart Paperweight that not only serves as a desktop accessory but also functions as a smart home/office automation controller. The core objective is to integrate a microcontroller (ESP32) with a relay circuit to allow wireless control of electrical appliances such as lights and fans. This system will enable users to operate appliances remotely using predefined commands sent from the smart paperweight itself or via wireless communication.

The smart paperweight will be designed to include embedded control buttons or sensors for direct interaction, as well as a wireless module for remote access using Wi-Fi or Bluetooth. It will connect to relays that can turn devices on or off and, in the case of fans, even control their speed using a motor driver circuit. The ESP32 microcontroller will act as the central processing unit, receiving inputs from the paperweight interface or external wireless signals, processing those commands, and activating the appropriate output.

The proposed system will be powered by a stable DC source and designed to maintain compactness, making it suitable for office desks or bedside tables. Additionally, the system will be designed for energy efficiency, user convenience, and safety, with proper insulation and protection for electrical components. A major focus will be on designing a clean and durable housing for the paperweight that blends functionality with aesthetics.

Through this project, we aim to demonstrate the practical application of embedded systems and IoT in daily life. The system will offer a cost-effective and user-friendly solution for automating basic home or office appliances, helping to bridge the gap between traditional control systems and modern smart automation technology.

OVERALL DESCRIPTION

LANGUAGE USED IN PROJECT:

In our project Arduino language is used.

HISTORY OF PROJECT:

Project Initiation (Month 1-2):

The idea for a IoT Based Smart Paperweight is conceived either by students themselves or through brainstorming sessions with faculty advisors.

Initial discussions focus on defining project objectives, scope, feasibility, and potential challenges.

A project team is formed, comprising students with backgrounds in computer engineering.

Research and Planning (Month 2-3):

The team conducts comprehensive research on existing IoT Based Smart Paperweight, Automation technology, microcontrollers, batteries, and other necessary components.

They explore different design options, considering factors like cost, complexity, functionality, and available resources.

A detailed project plan is developed, outlining tasks, timelines, budget estimates, required materials, and milestones.

Prototyping and Development (Month 4-7):

The team begins prototyping the Paperweight, starting with basic functionalities like Bluetooth connectivity and basic Component (if applicable). We selected and integrate the necessary hardware components (microcontroller, Bluetooth module, PCB , Acrylic) and write code for communication protocols.

Iterative testing and debugging are conducted throughout this phase to identify and address technical issues, refine algorithms, and improve overall performance.

Integration and Testing (Month 8-9):

The individual components of the IoT-based Smart Paperweight are integrated into a unified system, ensuring seamless hardware and software interaction. Comprehensive testing is conducted to verify the functionality of sensor accuracy, Wi-Fi/Bluetooth connectivity stability, and remote control responsiveness. The smart paperweight undergoes real-world testing in different environments to evaluate its performance under varied conditions, identifying potential weaknesses or limitations for further optimization.

Refinement and Optimization (Month 10-11):

Based on testing feedback and user experience observations, the team refines and optimizes the IoT-based Smart Paperweight to enhance its performance, user interface, response time, and overall usability. Advanced features such as adaptive learning, enhanced automation, or integration with external APIs may be implemented for extended functionality. During this phase, special attention is given to power efficiency, compact size, and aesthetic design to improve the product's practicality and appeal.

Documentation and Presentation (Month 12):

The team prepares comprehensive documentation covering the project objectives, system design, technical specifications, implementation challenges, testing results, and key lessons learned. A final presentation is created to showcase the IoT-based Smart Paperweight's features, capabilities, and potential applications to a panel of evaluators, faculty members, and industry experts. The project is presented as part of the diploma program's capstone showcase, where students demonstrate their achievements and answer questions from peers and evaluators.

Diploma Project Evaluation (Month 13):

The completed IoT-based Smart Paperweight project is presented to faculty members and evaluators as part of the diploma project assessment. The presentation includes a demonstration of the smart paperweight's functionality, followed by a Q&A session, allowing evaluators to assess the project’s effectiveness, innovation, and practical applications.

OBJECTIVES OF PROJECT

The objective of this project is to develop an IoT-enabled smart paperweight that enhances home and office automation by intelligently controlling a fan and light. This system aims to improve energy efficiency, user convenience, and smart automation by integrating sensors, IoT connectivity, and remote control features.

The smart paperweight will function as a multi-purpose device, ensuring automatic and adaptive environmental control based on real-time data. By leveraging wireless communication and automation technologies, this project provides a compact, cost-effective, and user-friendly solution for smart living spaces.

The primary objectives of this project are:

- **Automation of Fan and Light Control:** Enable automatic switching based on environmental conditions to enhance comfort and energy efficiency.
- **Remote Accessibility:** Allow users to control fan speed and light brightness from anywhere using a mobile app.
- **Energy Efficiency**: Optimize power consumption by ensuring the fan and light operate only when necessary.
- **User Personalization:** Implement learning algorithms to adapt to user preferences over time.
- **Smart Scheduling:** Provide a scheduling feature to set predefined operation times.
- **Compact and Multi-functional Design:** Serve as a decorative yet practical paperweight with embedded smart automation features.

SYSTEM REQUIREMENTS:

1. **Software requirements:**
    - Arduino IDE
    - Proteus
    - CorelDraw
    - Fusion 360
2. **Hardware requirements:**
    - PCB
    - Drilling
    - Soldering
    - Battery
    - Acrylic

MODULES:

In our project only one modules is used. The module is Bluetooth module.



FIG : BLUETOOTH MODULE

A Bluetooth module is a wireless communication device that enables short-range data transmission between electronic devices, typically within a range of 10 meters. It uses radio waves to establish a secure and reliable connection, making it suitable for applications like wireless headphones, speakers, and IoT devices.

ADVANTAGES AND DISADVANTADES:

Advantages	Disadvantages
Energy Efficient: Operates only when required, reducing power consumption.	**Internet Dependency**: Requires a stable internet connection for remote access and cloud-based features.
Convenient and Smart: Remote and voice control enhance usability.	**Initial Cost**: Higher cost compared to traditional paperweights due to electronic components.
Multi-Purpose: Serves as both a decorative paperweight and an automation hub.	**Technical Complexity**: Requires setup, configuration, and maintenance, which may not be user-friendly for all.
Compact & Wireless: Eliminates the need for manual switches.	**Power Consumption**: While optimized, it still consumes power compared to a traditional paperweight.
User-Centric Customization: Adapts to individual preferences and schedules.	**Security Concerns**: IoT devices are vulnerable to cyber threats if not properly secured.

APPLICATION AREAS:

- **Education:**
  Smart learning environments where the IoT-based smart paperweight can be used to monitor classroom conditions and automate lighting and cooling.
  Teaching aid for IoT concepts, allowing students to learn about sensor integration, automation, and wireless communication.

- **Healthcare:**
  Smart monitoring in hospitals to adjust room temperature and lighting automatically for patient comfort.
  Assisting elderly or disabled individuals by automating environmental adjustments based on personal preferences.

- **Customer Service:**
  Used in office reception areas to control lighting and fan settings automatically, improving energy efficiency and user comfort.
  Smart meeting rooms where conditions adjust automatically before scheduled meetings.

- **Home Automation:**
  Smart homes where the paperweight acts as an automation hub, controlling lights, fans, and other connected appliances.
  Personalization of indoor environments based on user habits, schedules, and voice commands.

- **Entertainment:**
  Mood lighting and cooling adjustments while watching movies or listening to music.
  Interactive desk accessory that responds to environmental changes with customizable LED lighting and smart notifications.

- **Workspaces & Offices:**
  Energy-saving solution by automating lighting and cooling in workspaces when a desk is in use.
  Productivity enhancement through custom settings that adapt to user preferences, reducing manual adjustments.

TESTING :

Testing is a process of executing a program with the aim of identifying and fixing errors. To ensure the IoT-based Smart Paperweight performs efficiently, rigorous testing is conducted at multiple stages. Successful testing ensures error-free operation, reliability, and seamless user experience.

Types of Testing:

1. **Unit Testing**
   - Focuses on testing individual components of the smart paperweight, such as sensor readings, connectivity modules, and automation algorithms.
   - Conducted by developers using sample inputs to verify expected outputs.
2. **Unit Testing:**
   - Focuses on testing individual components such as sensors, microcontrollers, and software modules.
   - Each unit is tested separately using sample input and expected outputs to ensure proper functionality.
3. **Integration Testing:**
   - Ensures that all hardware and software components work together as a unified system.
   - Tests the interaction between the microcontroller, sensors, mobile app, and cloud services**.**
4. **Regression Testing:**
   - Performed when new features (such as additional automation functions) are added.
   - Ensures that existing functionalities like fan speed control and light automation are not affected by updates.
5. **Smoke Testing:**
   - A preliminary test to ensure the basic functionalities work before deeper testing.
   - Confirms that the paperweight can turn on, connect to Wi-Fi/Bluetooth, and interact with sensors.
6. **Alpha Testing:**
   - Conducted by developers in a controlled environment before releasing the product.
   - Ensures that the automation, remote control, and scheduling features function correctly.
7. **Beta Testing:**
   - Performed by a limited group of end-users in real-world environments.
   - Collects feedback on the usability, reliability, and effectiveness of the smart paperweight.

8. **System Testing:**
    - Evaluates the smart paperweight as a complete system under various conditions.
    - Includes security testing, recovery testing, stress testing, and performance evaluation**.**
9. **Stress Testing:**
    - Tests the paperweight under extreme conditions, such as high temperature variations or multiple user interactions.
    - Checks whether the system remains functional despite heavy loads.
10. **Performance Testing:**
    - Measures system response time, Wi-Fi/Bluetooth connectivity speed, and sensor accuracy**.**
    - Ensures the system operates efficiently with minimal latency.

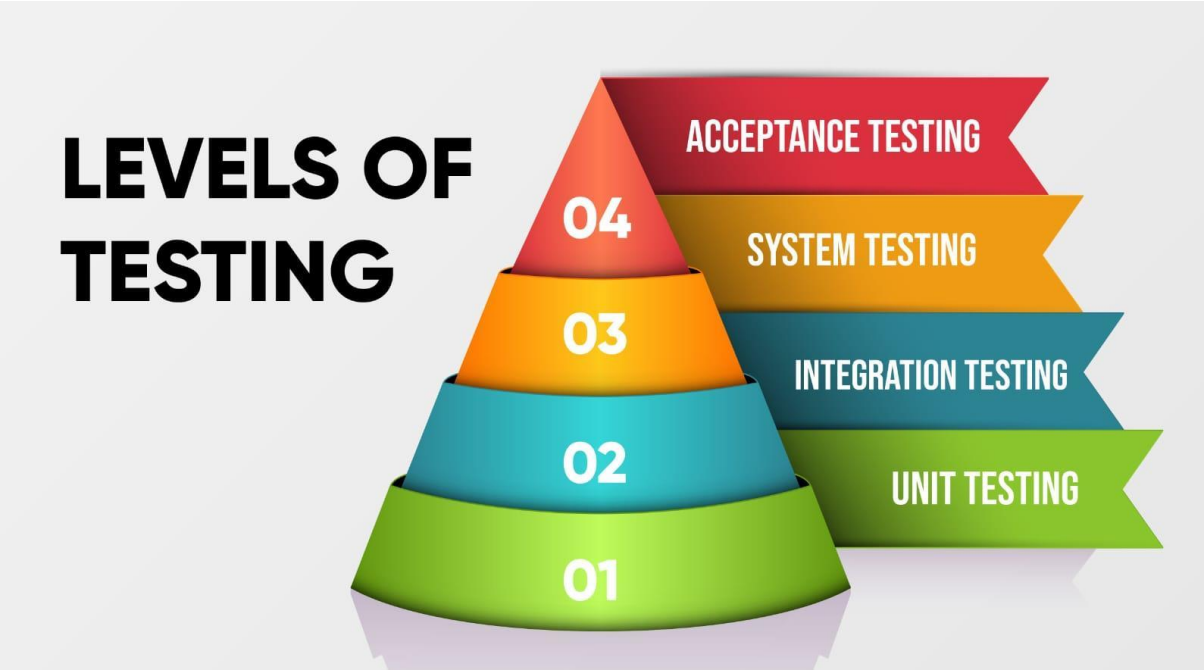
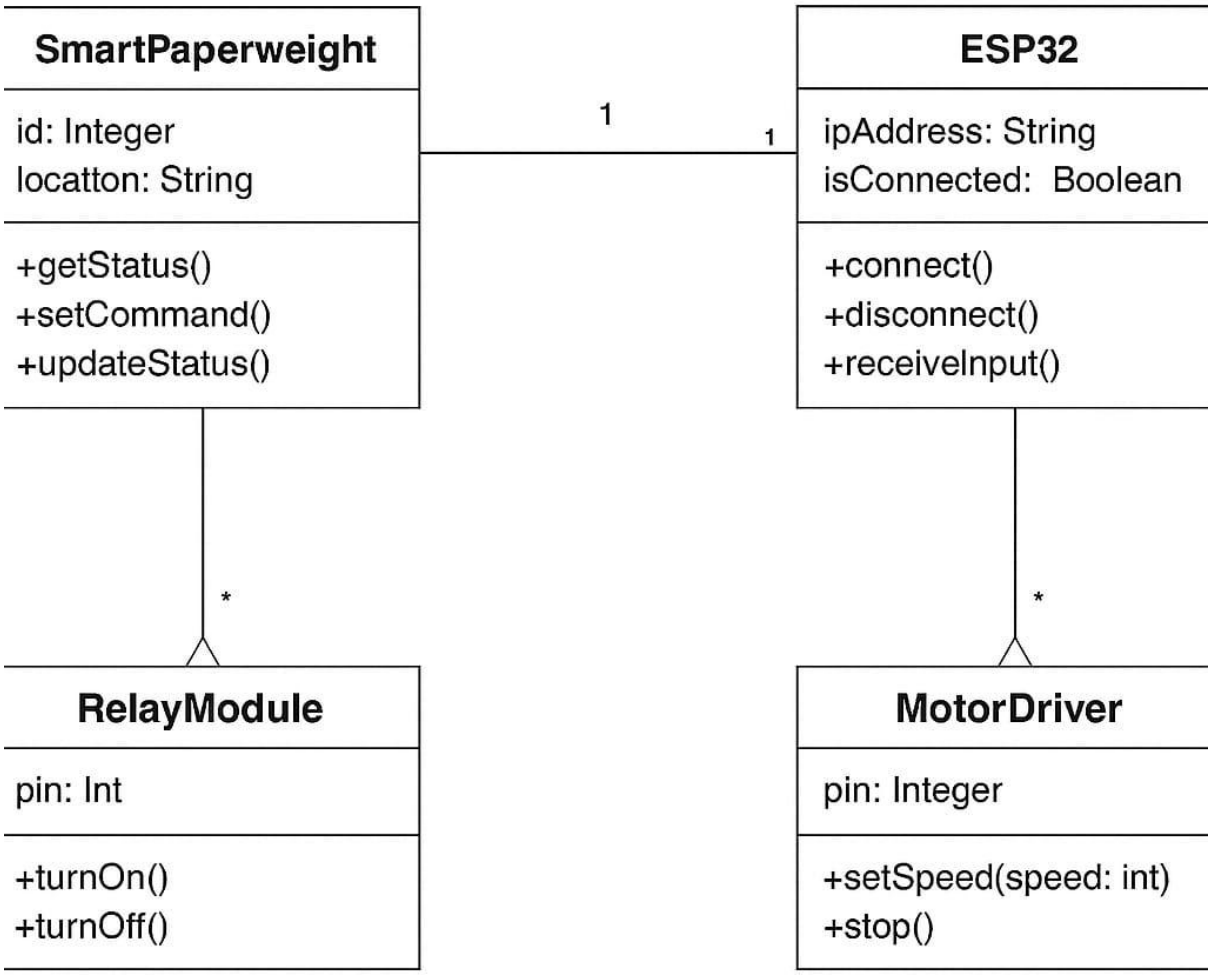
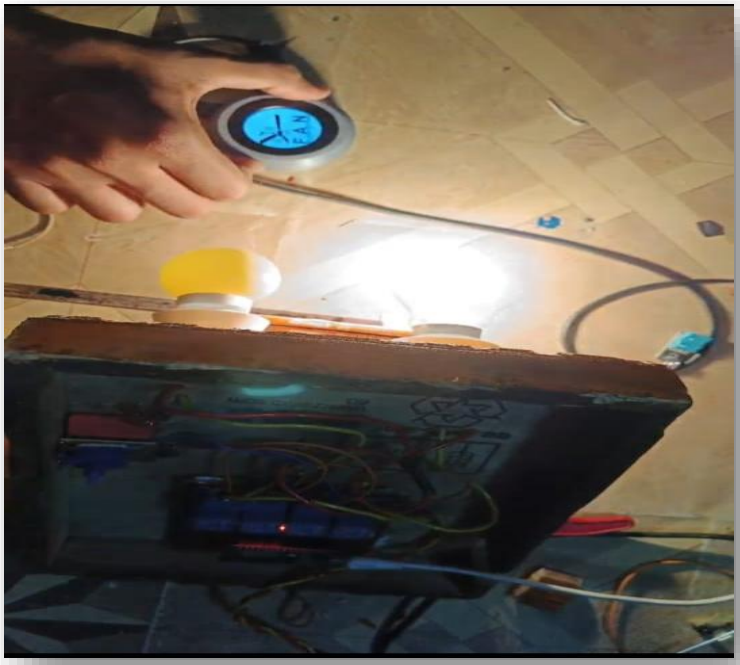


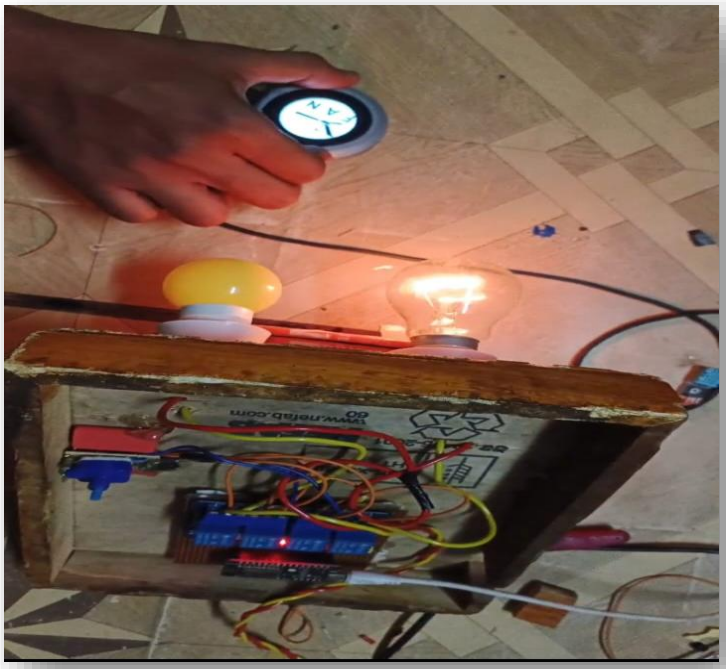
FIG : LEVEL OF TESTING

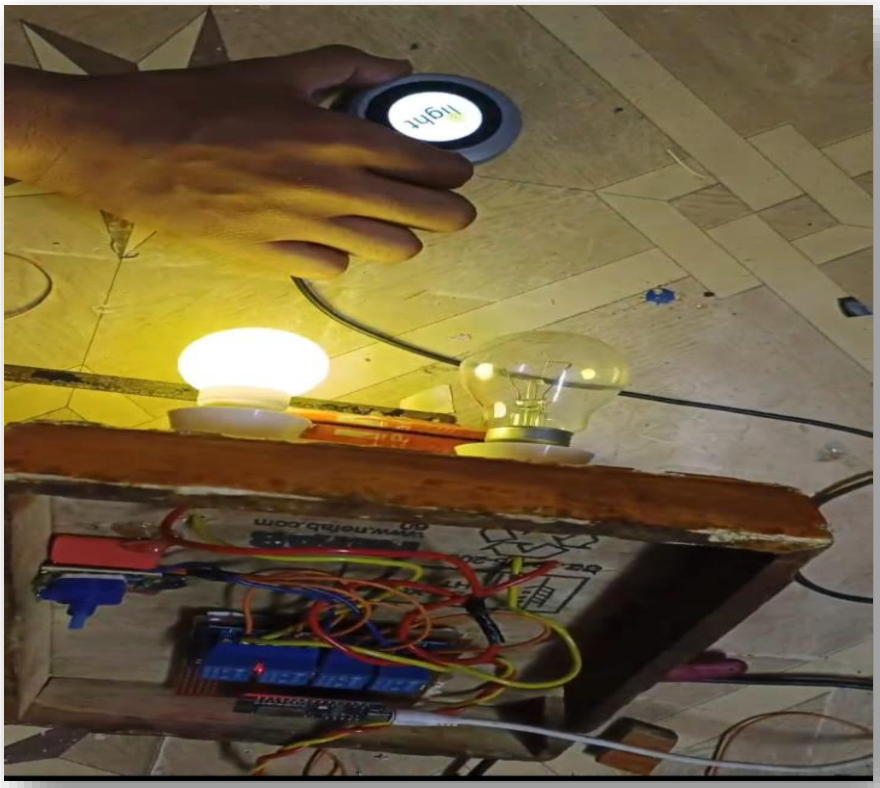
CLASS DIAGRAM:



RESULT






light

PROJECT SCOPE

The project scope for the IoT-based Smart Paperweight encompasses the development of both hardware and software components to create an efficient, automated system. The hardware development involves designing and assembling the smart paperweight, integrating essential elements such as microcontrollers, sensors, Wi-Fi/Bluetooth modules, relay modules, and power management components.

The software aspect focuses on programming the microcontroller to process sensor data, control the fan and light, implement IoT connectivity for remote access, and develop a user-friendly mobile app or web interface for seamless control. Additional features such as voice command integration, adaptive learning, and power monitoring enhance the system's efficiency and usability.

The project also includes thorough testing and debugging to ensure the reliability and functionality of all components. Various testing methods, including unit testing, integration testing, and performance evaluation, will be conducted to identify and resolve any technical issues.

Comprehensive documentation of project objectives, design specifications, technical implementation, and testing results will be maintained. A final presentation will showcase the features, capabilities, and real-world applications of the IoT-based Smart Paperweight, demonstrating its potential for smart home and office automation.

FUTURE SCOPE

IoT-based Smart Paperweight has the potential for further advancements and improvements, making it even more effective and versatile. Future developments may include:

- **AI-Based Predictive Automation:** Implementing machine learning algorithms to analyze user behavior and automate fan and light control based on daily routines.
- **Solar-Powered Operation:** Integrating solar energy harvesting to make the device energy-efficient and environmentally friendly.
- **Multi-Device Connectivity:** Expanding compatibility with multiple IoT devices and smart home ecosystems for enhanced automation.
- **Mobile App Enhancements:** Adding more features like custom automation rules, voice command customization, and cloud-based data storage for better user experience.
- **Gesture Control:** Introducing motion or gesture-based control for a more intuitive interaction with the paperweight.
- **Advanced Security Features:** Enhancing security through encrypted communication, user authentication, and multi-layer security protocols to prevent cyber threats.
- **Expanded Smart Office Applications:** Utilizing the smart paperweight for office automation, such as detecting user presence and adjusting workspace conditions accordingly.

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CONCLUSION

The IoT-based Smart Paperweight is an innovative solution that seamlessly integrates automation, convenience, and energy efficiency into everyday environments. By leveraging IoT technology, sensor-based automation, and wireless connectivity, it enhances user experience by automatically controlling fan speed and lighting based on environmental conditions. The device offers remote access, voice control, and adaptive learning, making it an essential addition to smart homes and offices.

Through rigorous design, testing, and optimization, the smart paperweight has been developed to provide energy savings, ease of use, and improved workspace comfort. Its compact and multi-functional design makes it not only a practical accessory but also an effective automation tool.

As technology evolves, future enhancements such as AI-based predictive control, gesture recognition, and expanded smart home integration will further improve its functionality. The IoT-based Smart Paperweight represents a step toward more intelligent and user-friendly automation, reinforcing the importance of IoT in modern living and working spaces.

The IoT-based Smart Paperweight demonstrates the practical integration of IoT in daily objects. By automating fan and light control with remote access and voice commands, it enhances convenience and energy efficiency. Future enhancements could include AI-based predictive control and solar-powered operation to further optimize functionality.

ACHIEVEMENT:

1.DIPEX Certificate:



2.INTERCOLLEGE COMPETITION:



K.E. Society's
RAJARAM BAPU INSTITUTE OF TECHNOLOGY
(POLYTECHNIC), LOHAGAON, PUNE

CERTIFICATE
Of
Participation

THIS CERTIFICATE IS PRESENTED TO:

Mr / Ms : Monal Sharad Meshram

On project : IoT Based Smart Paper Weight

Has Actively participated in
MECH-TECH Project Exhibition & Competition
Organized by :- Department of Mechanical Engineering & MESA
Conducted in 27th March 2025

Mr. S. S. Dabhade
Vice President

Mr. H.V. Nikalje
President

Mr. A.A. Satam
HoD

Dr. K.H. Munde
Principal

3.INTRACOLLEGE COMPETITION:



Akhil Bharatiya Maratha Shikshan Parishad's

Anantrao Pawar College of Engineering & Research,Pune

Accredited By NAAC With 'A' Grade , NBA- Civil & Mechanical

Under Mentorship of COEP Technological University

Sr. No.103,Shahu College Campus Parvati, Near Swargate ,Pune

College DTE Code:6794

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This is to Certify that

Ms. Monal Meshram from RITP, Computer Engineering Department, has Participated in Online 'APCOER TECHNOTHON-2025' Project/Paper/Start-up Competition held on 25th & 26th March, 2025.

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